Date : 06.07.2023
Thursday
2:00 pm to
3:30 pm.

Theory of Computation
21IS44
CIE-1 Scheme & Solution

1a] Distinguishable / Not Equivalent states:

$\delta(p,w) \in F$ & $\delta(q,w) \notin F$ or vice versa - 1M

Indistinguishable / Equivalent states:

Both $\delta(p,w)$ & $\delta(q,w)$ are final or non-final. - 1M

* Remove unreachable states if any - NO.

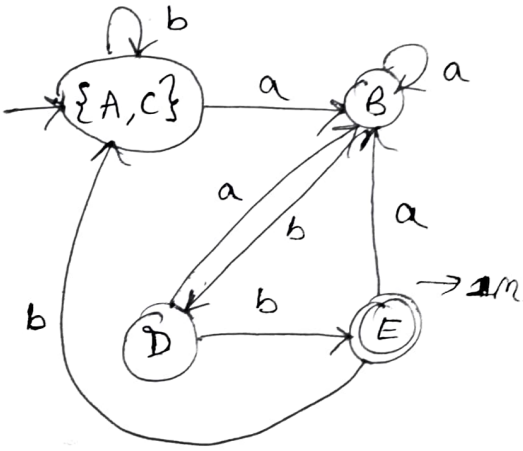
	a	b
* → A	B	C
B	B	D
C	B	C
D	B	E
* E	B	C

→ 1M

	A	B	C	D
B	X 3			
C	•	X 3		
D	X 2	X 2	X 2	
E	X 1	X 1	X 1	X 1

→ 3M

⇒



→ 1M

Reduced DFA

(A,C) → Not marked
↳ Indistinguishable / Equivalent pair → 1M

1b] Steps: ① For every production $A \to aB$, make $\delta(A, a) = B$ make a edge labelled a from A to B.

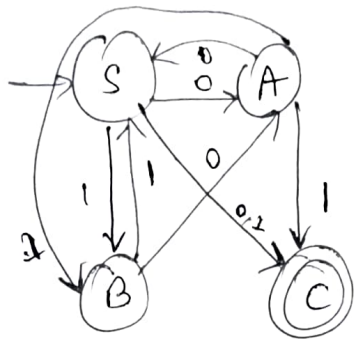
② For every production $A \to a$, make $\delta(A, a) =$ Final st

③ For every production $A \to \epsilon$, make $\delta(A, \epsilon) = A$ & A will be final state.

* No of states in the automata will be equal to no. of non-terminals plus one.

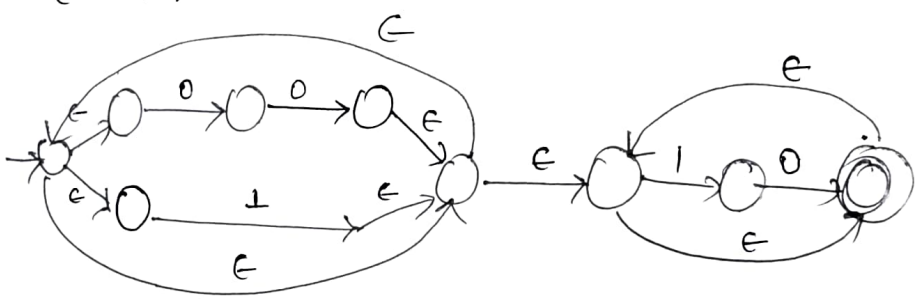
1M

Soln



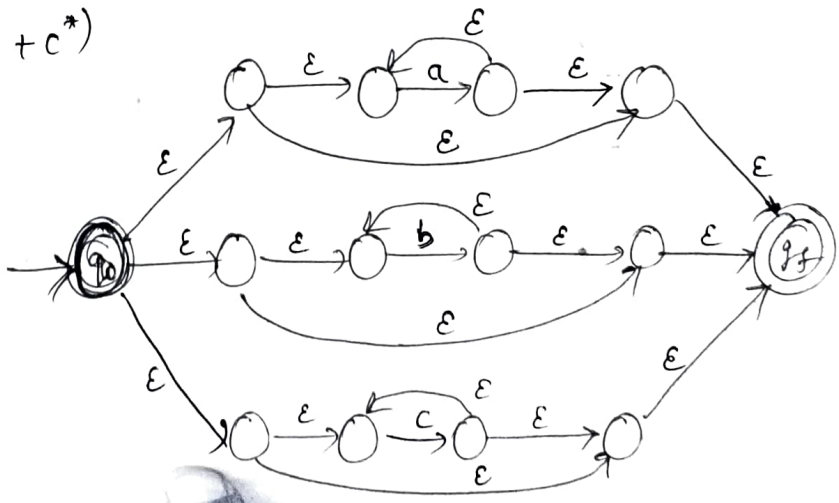
- 2M

2a] i) $(00+1)^* (10)^*$



2.5 x 2 = 5M

ii) $(a^* + b^* + c^*)$



Applications of R.E

* R.E in Unix
* Lexical Analysis
* Finding Pattern in text

2M

Algebraic laws of RE

* Associativity
* Commutativity
* Identity law
* Idempotent law
* Distributive law

Any 3

with Ex Explain

3M

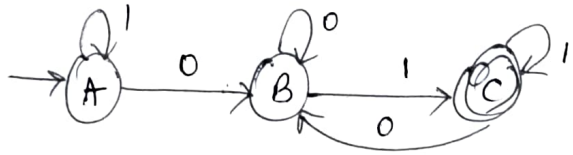
3 a] A grammar is said to be left linear if all productions are of the form:

$$A \to Bx$$
$$A \to x$$

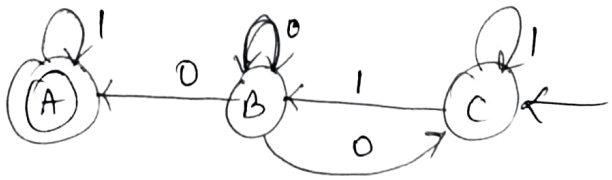
1M

where $A, B \in V$ & $x \in T^*$.

Given



Reverse the DFA



1M

RLG

$C \to 1C \mid 1B$
$B \to 0A \mid 0C \mid 0B$
$A \to 1A \mid \epsilon$

1M

$\xrightarrow{\text{Reverse}}$

$C \to C1 \mid B1$
$B \to A0 \mid C0 \mid B0$
$A \to A1 \mid \epsilon$

2M

3b] P.L statement:

Let L be regular language. Then there exists a constant n (which depends on L) such that for every string w in L such that $|w| \geq n$, we can break w into three strings $w = xyz$ such that

1. $y \neq \varepsilon$
2. $|xy| \leq n$
3. for all $k \geq 0$, the string xy^kz is also in L.

1M

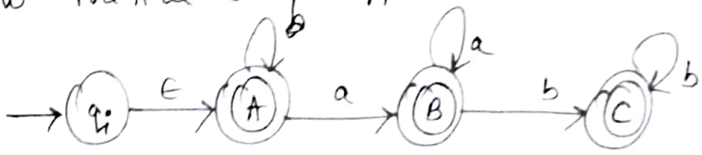
prove that $L = \{ww^R \mid w \in (0+1)^*\}$ is not regular 4M.

4a] Intersection - 1M
Homomorphism - 2M
Complementation - 1M
} R.L closure properties. proof.

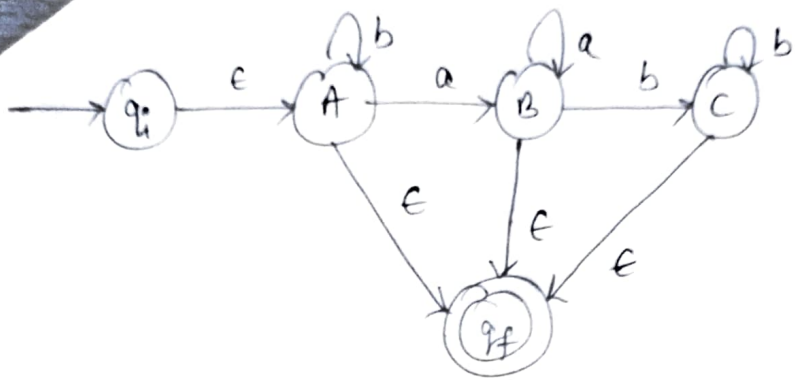
4b] In a given FA, D is dead state. Hence eliminate D & its associated edges.

Since initial state A has an incoming edge (self loop) create a new initial state q_i

i.e

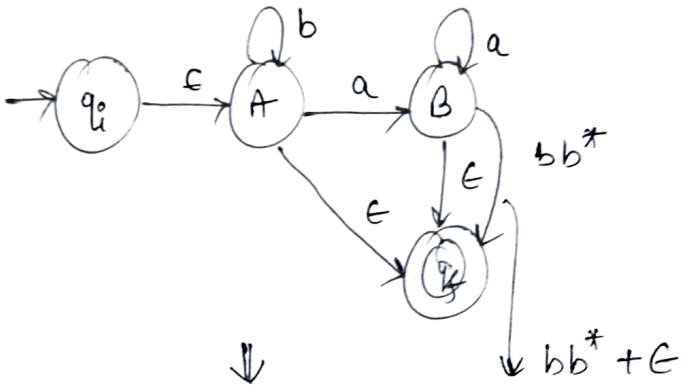


There exists multiple final states so convert them into a single final state.

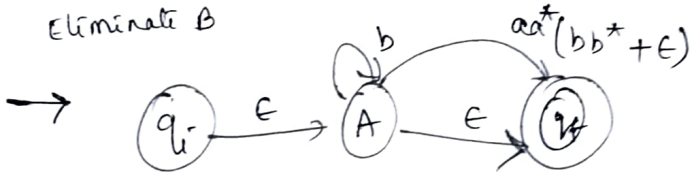


Now start eliminating the intermediate states

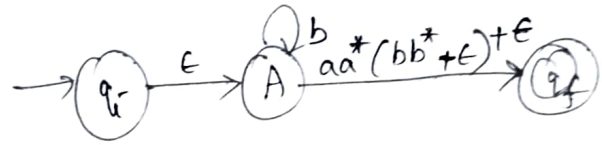
Eliminate C.



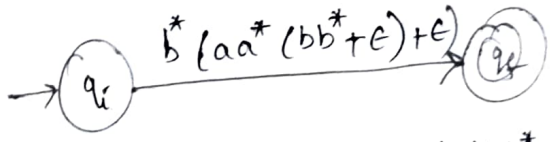
Eliminate B



i.e



Eliminate A



$\therefore$ RE is $b^* (aa^* (bb^* + \epsilon) + \epsilon)$ //.

5a] Any two decision properties with Ex.

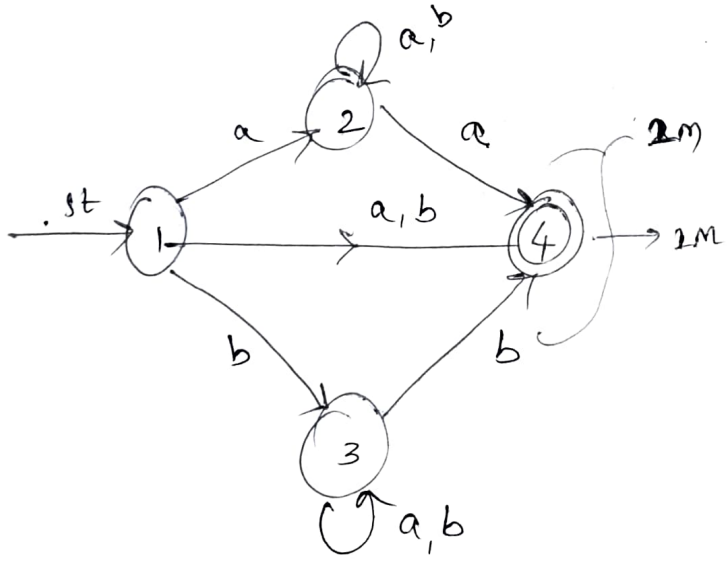
* Finiteness problem
* Emptiness
* Membership
* Equivalence.

2.5 * 2 = 5

5b] i) $0^*10^*10^*10^*$

ii) $((0+1)(0+1))^*$

iii) $1^*01^*01^*$

1 * 3 = 3M.

5c]